Salma **Mohammadzadeh**

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Summary

Computer Engineering student at Politecnico di Torino with proven experience across data pipelines, computer vision, and systems programming — from building ETL pipelines with PostgreSQL to deploying YOLOv8 object detection for a competitive robotics team (4th / 126 teams, ERC 2025). Comfortable across the full stack from data ingestion to visualization. Looking to contribute to meaningful engineering work through an internship.

Education

Politecnico di Torino — *Torino, Italy*
BACHELOR OF COMPUTER ENGINEERING — *2022 - Present*

- Relevant Coursework: Introduction to Databases, Cybersecurity, Computer Science & Programming Techniques

Experience

DIANA Student Robotics Team — *Politecnico di Torino*
COMPUTER VISION & ROBOTICS ENGINEER — *2023 - Present*

- 4th place out of 126 teams — European Rover Challenge 2025
- Built a computer vision pipeline for geological probe detection using YOLOv8, personally handling image collection, annotation, dataset organization, and model training in Python
- Prepared the training dataset through a mix of manual collection, sourced imagery, and augmentation techniques
- Supported rover arm kinematics simulation using MATLAB and Simulink alongside the main CV workstream

Python MATLAB Simulink YOLOv8 OpenCV CVAT Roboflow

Projects

SpaceX Launch Data Pipeline — *2026*
GITHUB.COM/SALMAZE/SPACEX-PIPELINE

- Designed and built a 4-stage ETL pipeline ingesting 205 SpaceX launches from a public REST API, transforming and aggregating the data with Pandas, and persisting it across two relational tables in a Dockerized PostgreSQL 15 database
- Delivered a live Streamlit dashboard querying Postgres directly, displaying KPI metrics and per-year launch success charts — entire stack spins up with a single command via Docker Compose

Python Pandas PostgreSQL Docker Streamlit

Triple DES Encryption — *2025*
GITHUB.COM/SALMAZE/CRYPTOGRAPHY-TRIPLE-DES

- Implemented full 3DES encryption and decryption from scratch in pure Python with zero external libraries, building all core cryptographic primitives: S-boxes, Feistel structure, Initial/Final Permutations, and 48-round key scheduling across three independent keys
- Built for a university Cybersecurity course; includes ECB mode, optional PKCS-style padding, and a written technical report

Skills

Languages	Persian (Native) | English (Native) | Azerbaijani (C1) | Italian (A2)
Programming	Python | SQL | C | MATLAB
Tools & Platforms	Git | GitHub | Docker | PostgreSQL | Streamlit | SQLAlchemy | Simulink
ML & Data	YOLOv8 | EasyOCR | Pandas | OpenCV | CVAT | Roboflow
Concepts	REST APIs | ETL Pipelines | Object Detection | Cryptography | Prompt Engineering

Certifications

Sep 2024	**ERC Space & Robotics Industry Standard**, European Rover Challenge
Sep 2025	**SpaceCert**, European Space Foundation
Ongoing	**Building with the Claude API**, Anthropic